When Machine Learning Beats Persistence: A Leakage-Free Streamflow and Drought Benchmark for the Tropical Andes

Richar Andre Vilca-Solorzano, Renato Quispe-Vargas and Dina Maribel Yana-Yucra

Faculty of Statistical and Computer Engineering, Universidad Nacional del Altiplano, Av. Floral 1153, Puno 21001, Peru

Corresponding author: R. A. Vilca-Solorzano, 75521963@est.unap.edu.pe, Tel.: +51 952 190 381
R. Quispe-Vargas, 72535253@est.unap.edu.pe; D. M. Yana-Yucra, maribelbel201314@gmail.com

ORCID: R. A. V.-S. 0009-0003-2385-5263; R. Q.-V. 0009-0005-2367-7852;
D. M. Y.-Y. 0009-0003-6218-2735

Abstract Machine-learning studies of Andean streamflow rarely compare their models with persistence, and their validation designs can leak information from the test period. We built an open benchmark of 19 long-record gauges from the Andean Glacierized Catchment dataset in three regions: the Altiplano, including Puente Isla Cabanillas in the Lake Titicaca basin, the Cordillera Blanca, and the western and southern Andes. Daily discharge was combined with ERA5, MERRA-2 and CHIRPS precipitation and GloFAS v4 simulated discharge. Six learners were compared with persistence and climatology at 1-, 3- and 7-day leads under a calendar-continuous, embargoed protocol tuned on three validation blocks. Uncertainty was estimated with block-bootstrap intervals and Diebold–Mariano tests. The learners beat persistence at 16 of 19 gauges at one day and at 17 of 19 at seven days (median skill 0.108). Lower error did not improve drought detection: the median critical success index of drought days was 0.43 for persistence and 0.25 for the learners. A drought threshold calibrated on validation data raised it to 0.40, close to but still below persistence (0.45). Drought synchrony decayed over about 582 km.

Key words: Tropical Andes · Altiplano · Streamflow forecasting · Hydrological drought · Machine learning · Data leakage

1 Introduction

Rivers of the tropical Andes supply irrigation, hydropower and drinking water, but their flow is strongly seasonal. On the Altiplano, rivers such as the Coata–Cabanillas feed Lake Titicaca and depend on a short austral-summer wet season [8], and hydrological droughts are projected to become longer and more frequent [24]. Short-range forecasts are therefore useful, but gauges are sparse and records gappy.

Table 1 Data-driven streamflow studies in the Andes. Pers.: persistence reference; CI: uncertainty of skill. Scores as reported by the authors.

Study	Region	Gauges	Step, lead	Method	Pers.	CI	Reported score
Muñoz et al. 2018 [13]	Ecuador	1	4–24 h	RF	no	no	NSE 0.76–0.38
Contreras et al. 2021 [3]	Ecuador	1	4–24 h	RF	no	no	R^2 0.69–0.29
Merizalde et al. 2023 [12]	Ecuador	1	1–11 h	LSTM	no	no	NSE 0.93–0.67
de la Fuente et al. 2019 [6]	Chile	9	hourly, 3 d	LSTM	no	no	NSE ≈0.93–0.99
Núñez et al. 2023 [15]	Chile	1	1–4 months	RF	no	no	R^2 0.85–0.73
Teverovsky et al. 2023 [20]	Argentina	1	1 month	SVR	no[a]	no	NSE up to 0.90
Vásquez-R. et al. 2023 [21]	Peru	1	monthly	ANN	no	no	r^2 0.81
Lujano et al. 2023 [11]	Peru (Altiplano)	1	1 day	k-NN	no	no	NSE 0.979
This study	Peru, Chile	19	1–7 days	6 learners	yes	yes	skill 0.108 at 7 days

[a]Compared with seasonal ARIMA.

Machine learning (ML) is widely used to forecast streamflow [14]. At daily leads, however, discharge is so autocorrelated that persistence, $\hat{Q}_{t+h} = Q_t$, is hard to beat [16]. Reported gains can be inflated by weak references, by shuffled cross-validation that leaks temporal information [2, 22], and by giving the baseline and the learners different information at the time of issue. An earlier version of our own pipeline made the last error, and gave the learners discharge only up to $t-1$.

We measured how much ML adds to persistence for 1- to 7-day streamflow and drought forecasts in the Andes when no information leaks from the test period. The paper makes four contributions: (i) an open benchmark of 19 gauges in three hydroclimatic regions; (ii) a leakage-free protocol checked by automated tests; (iii) skill against persistence with confidence intervals, compared with earlier Andean studies; and (iv) a direct, calibrated evaluation of drought-day detection. Section 2 reviews related work, Sect. 3 describes the data and methods, Sect. 4 presents the results, Sect. 5 discusses them with their limitations, and Sect. 6 concludes.

2 Related Work

Data-driven runoff forecasting in the Andes includes random forests and LSTM networks for hourly flows in Ecuador [3, 12, 13], LSTM flood warning and random forests in Chile [6, 15], and support-vector regression in Argentina [20]. In Peru, neural networks and k-nearest neighbours were used in two basins, including the Ramis River on the Altiplano [11, 21]. Table 1 summarizes these studies. Seven of the eight used a single target gauge, none compared the models with persistence, and none reported an uncertainty estimate for the skill. For Peruvian agricultural data, blocked cross-validation showed that random folds overstate accuracy [22]. Drought forecasts also need an explicit identification step [19], so we score drought days directly.

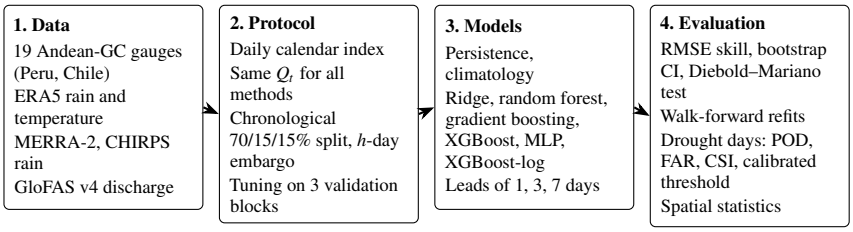


Fig. 1 Workflow of the benchmark. Code, tests and result tables are in the public repository.

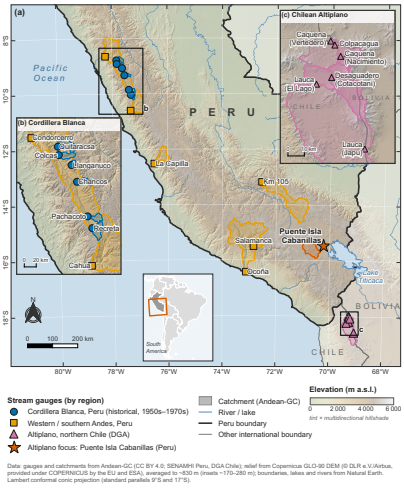


Fig. 2 The 19 gauges and their catchments by region. The focus gauge, Puente Isla Cabanillas, is highlighted.

3 Data and Methods

Data. Figure 1 summarizes the workflow. We used quality-controlled (QC) daily discharge from the Andean-GC dataset [1]. We kept every Peruvian gauge of the National Meteorological and Hydrological Service (SENAMHI) and every Chilean Altiplano gauge of the General Directorate of Water (DGA) with at least 5000 QC days. This gave 19 gauges: 7 on the Altiplano, 6 in the glacier-fed Cordillera Blanca, and 6 in the western and southern Andes (Fig. 2). The focus gauge was Puente Isla Cabanillas (2843 km^2, mean elevation 4471 m), which drains into Lake Titicaca. Daily precipitation and 2-m temperature came from ERA5 [9] at each catchment centroid through Open-Meteo [23]. MERRA-2 precipitation and GloFAS v4 discharge were added in a data-fusion test, and CHIRPS v2.0 precipitation (0.05°, averaged over each catchment, from 1981) [7] in a resolution test at the 13 gauges with records after 1981.

Forecasts and learners. A forecast issued on day t predicts Q_{t+h}, $h \in \{1,3,7\}$ days, from 27 predictors available at t. These were discharge lags (0 to 30 days), 7- to 30-day rolling means and standard deviations, precipitation lags and 7- to 90-day totals, temperature means, and the day of year. A lag of k days is exactly k calendar

days, and samples touching a gap were dropped. The references were persistence (same Q_t) and a day-of-year climatology. The six learners were Ridge regression, random forest, histogram gradient boosting, XGBoost, a two-layer perceptron (MLP), and an XGBoost trained on the change in $\log(1+Q)$ (XGBoost-log).

Protocol. Each record was split chronologically into 70% training, 15% validation and 15% test, with an h-day embargo between blocks. Hyper-parameters and the target form (level or change) were chosen on the mean root-mean-square error (RMSE) over three chronological blocks: the validation block and two earlier, equally long blocks of training. The chosen model was refit on training plus validation and scored once on test. Ridge was also run in a three-fold expanding walk-forward at every gauge. Automated tests check calendar offsets, the absence of future information and the embargo.

Scores. We report the Nash–Sutcliffe efficiency (NSE) and the skill $S = 1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{pers}}$. Its 95% confidence intervals (CIs) came from a moving-block bootstrap [10], and equal accuracy was tested with the Diebold–Mariano (DM) test [4]. A day was in drought when Q fell below the monthly 20th percentile of the fitting period [19]. Detection was scored with the probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI) [18]. Because RMSE-trained models are conservative near the threshold, we also calibrated the drought decision: a forecast signals drought when $\hat{Q} < k\,Q_{20}$, with k from 0.80 to 1.80 in steps of 0.05 chosen to maximize CSI on the validation block. The same calibration was applied to every method, including persistence. Drought synchrony was the correlation of a nonparametric 30-day standardized streamflow index (SSI-30) [5] between gauge pairs, fitted as $\rho(d) = c\,e^{-d/d_0}$; skill clustering was tested with Moran's I.

4 Results

Focus gauge. At Cabanillas (Table 2; test period 2021-09-22 to 2024-12-24), the validation-selected random forest reached NSE $= 0.912$ at one day, against 0.902 for persistence. This is a skill of 0.054 (95% CI -0.003 to 0.113; DM p 0.048), whereas our earlier, leaky pipeline had found a loss. At seven days, persistence fell to NSE $= 0.461$, and the selected Ridge regression gained 0.115 (95% CI 0.028 to 0.208); its walk-forward folds gave 0.166, 0.116, 0.114. Ridge beat persistence at all three leads with CIs above zero.

All gauges. The learner beat persistence at 16/19, 17/19 and 17/19 gauges at 1, 3 and 7 days, and its CI excluded zero at 11, 12 and 12 gauges (Fig. 3). The median skill grew from 0.051 to 0.080 and 0.108, because persistence degraded faster than the learners. Between one and seven days it rose in all three regions: from 0.060 to 0.103 on the Altiplano, from 0.033 to 0.097 in the Cordillera Blanca and from 0.053 to 0.130 in the western and southern Andes. Ridge, the most often selected learner, was skilful at 18/19 gauges at seven days, and 53 of 57 walk-forward folds had positive skill (median 0.103).

Table 2 Test scores at Puente Isla Cabanillas. Skill is relative to persistence, with 95% CI. †Learner selected on validation. CSI: drought-day detection at seven days.

	1-day lead		7-day lead		
Method	NSE	Skill [95% CI]	NSE	Skill [95% CI]	CSI
Persistence	0.902	–	0.461	–	0.61
Ridge	0.915	0.070 [0.01, 0.11]	0.578	0.115† [0.03, 0.21]	0.47
Random forest	0.912	0.054† [−0.00, 0.11]	0.528	0.064 [−0.02, 0.14]	0.59
Hist. gradient boosting	0.916	0.074 [0.01, 0.14]	0.437	−0.022 [−0.16, 0.12]	0.18
XGBoost	0.915	0.071 [−0.01, 0.13]	0.434	−0.025 [−0.17, 0.13]	0.56
MLP	0.932	0.169 [0.13, 0.20]	0.183	−0.231 [−0.51, −0.03]	0.25
XGBoost-log	0.916	0.077 [0.00, 0.13]	0.502	0.038 [−0.08, 0.18]	0.56

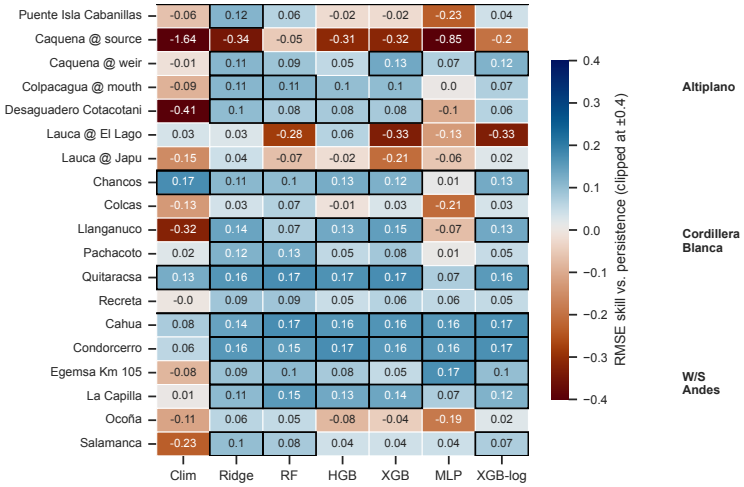


Fig. 3 Seven-day RMSE skill against persistence for every gauge and method. Boxed cells: 95% CI above zero. Gauges are grouped by region (top to bottom: Altiplano, Cordillera Blanca, western and southern Andes).

Drought detection. Lower RMSE did not give better detection of drought days (Fig. 4). At seven days, the median CSI was 0.43 for persistence and 0.25 for the learner, which beat persistence at only 3/19 gauges. The learners raised fewer false alarms (median FAR 0.38 vs. 0.39) but missed more drought days (POD 0.44 vs. 0.60). Calibration moved the learners towards more warnings (POD 0.68, FAR 0.53) and improved their CSI at 11 gauges, to a median of 0.40. Calibrated persistence reached 0.45 and remained better at 16 gauges.

Spatial statistics. The SSI-30 correlation of 104 gauge pairs decreased with distance (Spearman −0.56, p <0.001), with $d_0 = 582$ km (95% CI 330–1069 km), and was higher within regions than across them (median 0.28 vs. 0.09). Skill showed no spatial clustering (Moran's I =-0.012, $p = 0.540$), and no catchment attribute was significant after Holm correction (strongest: snow fraction, ρ =-0.45, $p = 0.056$).

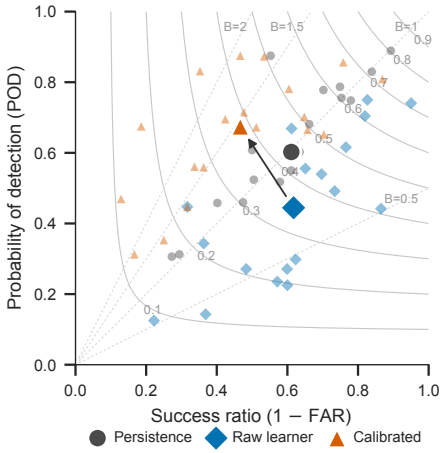


Fig. 4 Performance diagram [17] of seven-day drought-day detection. Small marks: gauges; large marks: medians. Curves: CSI (labelled on the diagonal); dotted lines: frequency bias B. The arrow shows the effect of calibrating the drought threshold.

5 Discussion

The protocol, not the model, changed our conclusions most: with equal information and calendar lags, a one-day loss at Cabanillas became a gain. The gains are modest (median RMSE 5–11% lower) but consistent, and Ridge was as good as tree ensembles [14].

Earlier Andean studies report high scores, often NSE above 0.9, but none was compared with persistence (Table 1). At Cabanillas, persistence alone reached NSE $= 0.902$ at one day and the best learner 0.912. For the Ramis River, a k-NN model with only the previous day's discharge reached NSE $= 0.965$, against 0.979 for the best model [11]. In both rivers most of the NSE comes from persistence, so a high NSE alone does not show skill.

Models trained on squared error smooth the recession limb near the drought threshold. Calibration closed most of this gap, but persistence remained the better drought reference, so warnings should be trained and verified on the drought decision itself [19]. More forcing data did not help. Adding MERRA-2 and GloFAS changed the seven-day Ridge skill by a median of -0.005. CHIRPS, which is 25 times finer and 2.8 times drier than ERA5 at these catchments, changed it by -0.009 and improved only 4 of 13 gauges. At short leads, recent discharge carries information that gridded rainfall does not add.

Limitations. Neither ERA5 nor CHIRPS resolves single convective storms on the Altiplano [8]. The Cordillera Blanca records predate MERRA-2 and GloFAS, so the fusion test covered 7 gauges only. Three selection blocks reduce, but do not remove, the effect of non-stationarity such as glacier retreat. Predictors used observed forcing up to t, not weather forecasts, and Andean-GC contains no Bolivian gauges.

6 Conclusion

We released an open, leakage-free benchmark for 1- to 7-day streamflow and drought forecasting at 19 Andean gauges. The learners beat persistence at 17/19 gauges at seven days (median skill 0.108). The gains did not carry over to drought-day detection; a calibrated threshold narrowed the gap but did not close it. Future work will add drought-aware training losses, operational weather forecasts and more Altiplano gauges.

Declarations

Data and code availability. Discharge: Andean-GC, doi:10.5281/zenodo.18035801. Code, tests, gauge list, tables and supplementary figures: `https://github.com/Andre031222/streamflow-drought-altiplano`. **Author contributions.** R. A.V.-S.: conception, software, analysis, writing; R.Q.-V.: validation; D. M.Y.-Y.: supervision. All authors approved the manuscript. **Competing interests.** None. **Use of AI.** Generative AI was used to improve the language; the authors reviewed the text and take full responsibility for it.

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